

NITHIN K

nkidang1@jh.edu | +14105001549 | [My Website](#) | [My LinkedIn](#)

SUMMARY

VLSI Design and Verification-focused ECE graduate from Johns Hopkins University with experience in **FPGA**, and **ASIC design** and functional verification. Skilled in **Verilog** and **SystemVerilog** languages with demonstrated work in AI hardware acceleration and in-memory computing. Seeking to contribute to the design and development of scalable, high-performance digital systems and next-generation computing solutions.

EDUCATION

JOHNS HOPKINS UNIVERSITY | Baltimore, MD

MSE in Electrical and Computer Engineering | Graduated - 05/2026 | 3.6/4.0

VELLORE INSTITUTE OF TECHNOLOGY, VIT | Tamil Nadu, India

B.Tech in Electrical and Electronics Engineering | Graduated - 07/2024 | 3.7/4.0

TECHNICAL SKILLS

- **RTL Design:** SystemVerilog, Verilog, Finite State Machines, Pipelining Strategies, Memory Hierarchy
- **Verification:** SystemVerilog Assertions, Universal Verification Methodology
- **Timing:** Static Timing Analysis, Clock Domain Crossing (CDC), Setup/Hold Analysis, Metastability Concepts
- **EDA Tools:** Cadence Xcelium, AMD Vivado, Cadence Genus, Cadence Innovus, Cadence Virtuoso, Intel Quartus
- **Programming:** C++, Python, MATLAB
- **Architecture:** RISC V, Mamba State Space Models, Transformer Models, In-Memory Compute

WORK EXPERIENCE

Teaching Assistant: First-Year ECE Design | Johns Hopkins University | Baltimore, MD | January 2026 – May 2026

- Mentored 30+ students over weekly lab sections and office hours, teaching circuit fundamentals and electrical design concepts.
- Graded homework, quizzes, lab reports, and communicated technical feedback.

Digital Design Researcher – AI Hardware | Johns Hopkins University | Baltimore, MD | May 2025 - January 2026 | [Project Link](#)

- Identified **latency bottlenecks** in the selective-scan computation of a Vision Mamba edge medical image classification model in **Python**.
- Designed **channel-parallel** 8-bit fixed-point fully-parameterized **Verilog RTL** for the selective-scan kernel - including pipelined datapath, memory, and state machines for control - reducing model latency by **9×** over the baseline.
- Implemented the accelerator on a **Kintex-7 FPGA**, meeting timing and resource utilization constraints in **AMD Vivado 2024.1**.
- [Published in IEEE Transactions on Biomedical Circuits and Systems](#)

NPU Verification Intern | STMicroelectronics | Uttar Pradesh, India | January 2024 - June 2024 | [Project Link](#)

- Architected a complete 6T SRAM-based analog in-memory compute architecture in **Simulink**, auto-generating synthesizable **Verilog RTL** for functional verification using **Cadence Xcelium**.
- Developed LFSR-based noise injection models to emulate $\pm 0.2\%$ bit-cell noise and ± 1 feature-level perturbations, stress-testing the architecture under **mixed-signal** operating conditions.

VLSI Design Intern | Maven Silicon Pvt. Ltd | Karnataka, India | May 2023 - July 2023 | [Project Link](#)

- Implemented **Verilog RTL** for the AHB interface, APB controller **Finite-State Machine**, burst-transfer logic (INCR4/WRAP4), and Verilog testbench for an AMBA AHB-to-APB bridge in **Intel Quartus Prime**.

TECHNICAL PROJECTS

RISC-V Processor based End-to-End Vision Mamba Accelerator | RND4IMPACT INC. | San Jose, CA | June 2026 - Present

- Designing microarchitecture for a **RISC-V AI SoC** for Vision Mamba acceleration, defining the processor-accelerator interface, memory-mapped control registers, DMA dataflow, and execution control.
- Implementing cycle-accurate behavioral models on **C++**, and parameterized **SystemVerilog RTL** for systolic GEMM engines, pipelined vector-processing datapaths, state machine control, nonlinear function blocks.

Evaluating State-Space Models for Memory-Efficient 3D Brain Tumor Segmentation | Johns Hopkins University | Baltimore, MD | January 2026 – May 2026 | [Project Link](#)

- Replaced Transformer encoder blocks in **Swin-UNETR** with **Vision Mamba modules** to evaluate state-space models as memory-efficient alternatives to Transformers for 3D medical image segmentation, using **PyTorch/Python**.
- Achieved 20% reduction in **peak GPU training memory**, 46.6% reduction in **peak inference memory**, and **comparable Dice and Hausdorff Distance** scores relative to Swin-UNETR baseline.

Design of 8-Bit & Binary Neural Network Accelerators | Johns Hopkins University | Baltimore, MD | August - December 2024 | [Project Link](#)

- Architected **SystemVerilog RTL** of 8-bit and binarized neural network **ASIC accelerators**, comprising Fully Connected MAC engines, memory hierarchies, address-generation logic, and control datapaths for MNIST classification.
- **Synthesized** the designs using **Cadence Genus**, demonstrating an **~8% reduction** in combinational logic for the BNN architecture at equivalent area.
- Completed **place-and-route** in **Cadence Innovus** and analyzed post-layout PPA across combinational, sequential, memory, and I/O components.

CERTIFICATIONS

- [Cadence SystemVerilog for Design and Verification v25.03 Certification \(Certified Feb 2026\)](#)
Integrated and verified a **VeriRISC CPU in SystemVerilog**, validating instruction execution, ALU functionality, memory operations, and program flow using directed diagnostic tests and a Fibonacci benchmark program.
- [Cadence SystemVerilog Assertions v5.2 Certification \(Certified March 2026\)](#)
Architected **assertion-based verification for an SPI interface**, capturing protocol sequences, transaction headers, suspend handling, illegal transaction detection, and coverage-driven validation using SVA.
- [Cadence Essential SystemVerilog for UVM v1.2.5rev3 Certification \(Certified June 2026\)](#)
Developed **Verification Components** to test a **4-port switch design**, covering **constrained-random** verification, virtual interfaces, sequencer, driver, and monitor subclasses through **Object-Oriented Programming** principles.
- [Cadence SystemVerilog Accelerated Verification with UVM v1.2.5rev2 \(Certified July 2026\)](#)
Developed a **UVM test to verify a YAPP packet-router DUT**, incorporating configurable UVC agents with drivers, sequencers, and monitors; multichannel sequence generators, TLM scoreboarding, test, and sequence libraries.

RELEVANT COURSEWORK

- Mixed Mode VLSI Systems
- CAD Design of Digital VLSI Systems (Graduate Coursework)
- FPGA Synthesis Lab (Graduate Coursework)
- VLSI Design (Undergraduate Coursework)
- Computer Architecture and Organization (Undergraduate Coursework)
- Object-Oriented Programming with C++ (Undergraduate Coursework)
- Generative AI for Medical Imaging and Computer Vision (Graduate Coursework)
- Foundations of Reinforcement Learning (Graduate Coursework)